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19-12-23

Discrete Structure

Imp
chinese remainder theorem

By Zout Identity:

$$\gcd(a,b) = sa + tb$$

's' ans 't' are by zout coeff

Example:

10 & 11

Sol:

Euclid theorem
$a = dq + r$

$$11 = 10(1) + 1$$

rearrange

$$1 = 11(1) + 10(-1)$$

By zout coeff 1 and -1

inverse = -1

* when gcd is 1 inverse exist and the num multiply with the lower value is the inverse

Example 2:

252 & 198

Solution:

$$252 = 198(1) + 54$$
$$198 = 54(3) + 36$$
$$54 = 36(1) + 18$$
$$36 = 18(2) + 0$$

gcd = 18

~~18 = 54(1) + 36(-1)~~

$$54 = 36(1) + 18$$

taking

$$18 = 54(1) + 36(-1)$$

using $198 = 54(3) + 36$

$$18 = 54(1) + (-1)[198(1) + 54(-3)]$$

$$18 = 54(1) + 198(-1) + 54(3)$$

$$18 = 54(4) + 198(-1)$$

using $252 = 198(1) + 54$

$$18 = [252(1) + 198(-1)](4) + 198(-1)$$

$$18 = 252(4) + 198(-4) + 198(-1)$$

$$18 = 252(4) + 198(-5)$$

by zout coeff 4 & -5

Example - 3

117 , 213

Solⁿ

$$213 = 117(1) + 96$$
$$117 = 96(1) + 21$$
$$96 = 21(4) + 12$$
$$21 = 12(1) + 9$$
$$12 = 9(1) + 3$$
$$9 = 3(3) + 0$$

gcd = 3

$$3 = 12(1) + 9(-1)$$
$$3 = 12(1) + (-1)[21(1) + 12(-1)]$$
$$3 = 12(1) + 21(-1) + 12(1)$$
$$3 = 21(-1) + 12(2)$$
$$3 = 21(-1) + (2)[96(1) + 21(-4)]$$
$$3 = 21(-1) + 96(2) + 21(-8)$$
$$3 = \cancel{21(-1)}\ 96(2) + 21(-9)$$
$$3 = 96(2) + (-9)[117(1) + 96(-1)]$$
$$3 = 96(2) + 117(-9) + 96(9)$$
$$3 = 96(11) + 117(-9)$$
$$3 = (11)[213(1) + 117(-1)] + 117(-9)$$
$$3 = 213(11) + 117(-11) + 117(-9)$$
$$3 = 213(11) + 117(-20)$$

By zoat coeff = 11 & -20

topic

linear congruence

Q: What is the solution of linear congruen

i) $55x \equiv 34 \pmod{89}$ —(i)

Sol:

general equation

$ax = b \pmod m$

take gcd of a & m

$\therefore$ gcd (55, 89)

$89 = 55(1) + 34$

$55 = 34(1) + 21$

$34 = 21(1) + 13$

$21 = 13(1) + 8$

$13 = 8(1) + 5$

$8 = 5(1) + 3$

$5 = 3(1) + 2$

$3 = 2(1) + 1$

$2 = 1(2) + 0$

Find inverse as gcd = 1

$1 = 2(-1) + 3(1)$

$1 = 3(1) + (-1)[5(1) + 3(-1)]$

$1 = 3(1) + 5(-1) + 3(1)$

$1 = 5(-1) + 3(2)$

$1 = 5(-1) + (2)[8(1) + 5(-1)]$

$1 = \cancel{8(-1)}\ 5(-1) + 8(2) + 5(-2)$

$1 = 8(2) + 5(-3)$

$1 = 8(2) + (-3)[13(1) + 8(-1)]$

$1 = 8(2) + 13(-3) + 8(3)$

$1 = 13(-3) + 8(5)$

$1 = 13(-3) + (5)[21(1) + 13(-1)]$

$1 = 13(-3) + 21(5) + 13(-5)$

$1 = 21(5) + 13(-8)$

$1 = 21(5) + (-8)[34(1) + 21(-1)]$

$1 = 21(5) + 34(-8) + 21(8)$

$1 = 34(-8) + 21(13)$

$1 = 34(-8) + (13)[55(1) + 34(-1)]$

$1 = 34(-8) + 55(13) + 34(-13)$

$1 = \cancel{3}\,55(13) + 34(-21)$

$1 = 55(13) + (-21)[89(1) + 55(-1)]$

$1 = 55(13) + 89(-21) + 55(21)$

$1 = \underline{89}(-21) + \underline{55}(34)$

$\Rightarrow$ inverse $= 34$

multiply inverse to eq 1

$(34)\ 55x \equiv (34)(34) \pmod{89}$

$1870x \equiv 1156 \pmod{89}$

~~if~~ b must be ~~less than~~ m

if $b > m$

$$a = dq + r$$

$$1156 = 89(12) + 88 \checkmark$$

$$x \equiv 88 \mod (89)$$

Example $19x \equiv 4 \mod (141)$ $\rightarrow$ 1

Sol $\gcd(141, 19) \Rightarrow$

$141 = 19(7) + 8$

$19 = 8(2) + 3$

$8 = 3(2) + 2$

$3 = 2(1) + 1$

$2 = 1(2) + 0$

Find inverse

$1 = 3(1) + 2(-1)$

$1 = 3(1) + (-1)[8(1) + 3(-2)]$

$1 = 3(1) + 8(-1) + 3(2)$

$1 = 8(-1) + 3(3)$

$1 = 8(-1) + (3)[19(1) + 8(-2)]$

$1 = 8(-1) + 19(3) + 8(-6)$

$1 = 19(3) + 8(-7)$

$1 = 19(3) + (-7)[141(1) + 19(-7)]$

$1 = 19(3) + 141(-7) + 19(49)$

$1 = 141(-7) + 19(52)$

$\Rightarrow$ inverse $= 52$

multiply inverse to eq 1

Discrete Structure

20-12-23

Chinese Remainder Theorem :

Example

$x \equiv 2 \pmod 3$ ⇒ when divided by 3 remainder is 2

$x \equiv 3 \pmod 5$

$x \equiv 2 \pmod 7$

Sol

$a_1 = 2, \ a_2 = 3, \ a_3 = 2$

$m_1 = 3, \ m_2 = 5, \ m_3 = \cancel{8}\ 7$

$M = (m_1)(m_2)(m_3)$

$M = 3 \times 5 \times \cancel{8}\ 7$

$M = 105$

$M_1 = \frac{M}{m_1} = \frac{105}{3}$

$\boxed{35 = 3(11) + 2}$

$= 35 \Rightarrow 35 \pmod 3$

$\Rightarrow 2 \pmod 3$

$M_2 = \frac{M}{m_2} = \frac{105}{5} = 21 \Rightarrow 21 \pmod 5 \Rightarrow 1 \pmod 5$

$\boxed{21 = 5(4) + 1}$

$M_3 = \frac{M}{m_3} = \frac{105}{7} = 15 \Rightarrow 15 \pmod 7 \Rightarrow 1 \pmod 7$

$\boxed{15 = 7(2) + 1}$

Find inverse

$2 \pmod 3$

$3 = 2(1) + 1$

$1 = 3(1) + 2(-1)$

inverse $= -1$

$\therefore \quad b_1 = -1$

For ear of M_2 and M_3 as mods and 7 are multiplsing with 1 so the invers for them is 1

$\therefore \quad b_2 = 1$

$b_3 = 1$

Chinese remainder therem

$$X = [a_1 M_1 b_1 + a_2 M_2 b_2 + a_3 M_3 b_3] \pmod{M}$$

$$X = [(2)(35)(-1) + (3)(21)(1) + (2)(15)(1)] \pmod{105}$$

$X = 23 \mod 105$

Conclusion:

23 is the smallest +ve integer that have a remaiheder 2 when divided by 3, a reamainder of 3 when divided by 5, a remainder of 2 when divide by 7.

Solve:

$x \equiv 1 \pmod{2}$

$x \equiv 2 \pmod{3}$

$x \equiv 3 \pmod{5}$

$x \equiv 4 \pmod{11}$

Sol:

$a_1 = 1$, $a_2 = 2$, $a_3 = 3$, $a_4 = 4$

$m_1 = 2$, $m_2 = 3$, $m_3 = 5$, $m_4 = 11$

$M = (m_1)(m_2)(m_3)(m_4)$

$= 2 \times 3 \times 5 \times 11$

$= 330$

$M_1 = \frac{M}{m_1} = \frac{330}{2} = 165 \Rightarrow 165 \pmod{2} \Rightarrow 1 \pmod{2}$

$165 = 2(82) + 1$

$M_2 = \frac{M}{m_2} = \frac{330}{3} = 110 \pmod{3} \Rightarrow 2 \pmod{3}$

$110 = 3(36) + 2$

$M_3 = \frac{M}{m_3} = \frac{330}{5} = 66 \Rightarrow 66 \pmod{5} \Rightarrow 1 \pmod{5}$

$66 = 5(13) + 1$

$M_4 = \frac{M}{m_4} = \frac{330}{11} = 30 \Rightarrow 30 \pmod{11} \Rightarrow 8 \pmod{11}$

$30 = 11(2) + 8$

$b_1 = 1$

$b_3 = 1$

For b_2

$$3 = 2(1) + 1$$

$$1 = 3(1) + 2(-1)$$

$$b_2 = -1$$

for b_4

$$\cancel{330 = 11(2)2 + 8}$$

$$11 = 8(1) + 3$$

$$8 = 3(2) + 2$$

$$3 = 2(1) + 1$$

using
Bezout

$$1 = 3(1) + 2(-1)$$

$$1 = 3(1) + (-1)[8(1) + 3(-2)]$$

$$1 = 3(1) + 8(-1) + 3(2)$$

$$1 = 3(3) + 8(-1)$$

$$1 = (3)[11(1) + 8(-1)] + 8(-1)$$

$$1 = 11(3) + 8(-3) + 8(-1)$$

$$1 = 11(3) + 8(-4)$$

$$\cancel{1 = 11(3) + (-4)[330(1) + 11(-2)]}$$

$$\cancel{1 = 11(3) + 330(-4) + 11(8)}$$

$$b_4 = -4$$

$$x = [a_1 M_1 b_1 + a_2 M_2 b_2 + a_3 M_3 b_3 + a_4 M_4 b_4] \bmod M$$

$$x = [(1)(165)(1) + (2)(110)(-1) + (3)(66)(1) + (4)(30)(-4)] \pmod{330}$$

$$x = -337 \pmod{330}$$

$$-337 = 330(-2) + 323$$

$$x \equiv 323 \pmod{330}$$

Q3

$$x \equiv 4 \pmod 5$$
$$x \equiv 3 \pmod 7$$
$$x \equiv 5 \pmod 9$$

Sol

$a_1 = 4, \quad a_2 = 3, \quad a_3 = 5$

$m_1 = 5, \quad m_2 = 7, \quad m_3 = 9$

$M = (m_1)(m_2)(m_3)$

$M = 5 \times 7 \times 9$

$M = 315$

$M_1 = \dfrac{M}{m_1} = \dfrac{315}{5} = 63 \Rightarrow 63 \pmod 5$

$\Rightarrow 3 \pmod 5$

$M_2 = \dfrac{M}{m_2} = \dfrac{315}{7} = 45 \Rightarrow 45 \pmod 7$

$\Rightarrow 3 \pmod 7$

$M_3 = \dfrac{M}{m_3} = \dfrac{315}{9} = 35 \Rightarrow 35 \pmod 9$

$\Rightarrow 8 \pmod 9$

For b1

$$5 = 3(1) + 2$$

$$3 = 2(1) + 1$$

using by zout

$$1 = 3(1) + 2(-1)$$

$$1 = 3(1) + (-1)[5(1) + 3(-1)]$$

$$1 = 3(1) + 5(-1) + 3(1)$$

$$1 = 3(2) + 5.(-1)$$

$$b_1 = 2$$

For b2

$$7 = 3(2) + 1$$

using by zout

$$1 = 7(1) + 3(-2)$$

$$b_2 = -2$$

For b3

$$9 = 8(1) + 1$$

$$1 = 9(1) + 8(-1)$$

$$b_3 = -1$$

$$X = [a_1 M_1 b_1 + a_2 M_2 b_2 + a_3 M_3 b_3] \bmod M$$

$$X = [(4)(63)(2) + (3)(45)(-2) + (5)(35)(-1)] \pmod{315}$$

$$X = 164 \pmod{315}$$

2-1-24 Discrete Structure

Algebraic Structure :

1) Semi Group — 2 properties
2) Monoids — 3 //
3) Group — 4 //
4) Ring — 5 //
5) Fields — 6 //

binary operations

① under Addition
② under Multiplication

Example :

$S = \{a, b, c, d\}$

binary operation $*$ is given by the table.

$*$	a	b	c	d
a	a	c	d	a
b	a	b	c	d
c	a	b	a	c
d	a	b	b	b

Sol

1) commutative

$a * b = b * a$

$c \neq a$

2) Associative

$$a * (b * c) = (a * b) * c$$

$$a * c = c * c$$

$$d \neq a$$

$\longleftrightarrow$

Groupp

let $(G, *)$ be an algebraic system where $*$ is binary operation on G. Then $(G, *)$ is a group if the following conditions are satisfied

1) G is closed under $*$
2) $*$ is associative
3) there exist identity element $e \in G$ for operation $*$
4) Each element $x \in G$ has an inverse $x^{-1} \in G$ w.r.t $*$

Example :

Consider the set $G = \{1, -1\}$ and let the binary operation defined on G be the multiplication than $(G, \cdot)$ is a group

Sol :

$\circ$	1	-1
1	1	-1
-1	-1	1

1) G is closed (all outputs belong to G)

2) ~~A~~ ~~G~~ Associative

$$A(B \cdot C) = (A \cdot B) \cdot C$$

let $C = A$

$$A \cdot (B \cdot A) = (A \cdot B) \cdot A$$

$$(1) \cdot (-1)(1) = (1)(-1)(1)$$

$$-1 = -1$$

G is Associative.

3) Identity element ~~e~~ $= 1$

4) Inverse

$$(1)^{-1} = 1$$

$$(-1)^{-1} = -1$$

$\Rightarrow (G, \cdot)$ is a group.

* If a group is commutative it is called abelian group

Discrete Structure 03-01-24

Groups Continue

Example:

$G = \{1, -1, i, -i\}$ under '.'

show that $(G, \cdot)$ ~~be~~ a Group

Sol:

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

1) G is closed
(all outputs $\in G$)

2) Associative

$a \cdot (b \cdot c) = (a \cdot b) \cdot c$

$1 \cdot (-1 \; i) = (1 \cdot -1) \cdot i$

$1 \; (-i) = -1 \cdot i$

$-i = -i$

G is Associative

3) Identity element = 1

4) Inverse

$(1)^{-1} = 1$

$(-1)^{-1} = -1$

$(i)^{-1} = -i$

$(-i)^{-1} = i$

$\Rightarrow (G, \cdot)$ is a group

next class

Quiz 3

Assignment 3

Example : 02

let $G\bar{5} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ be the set of modulo 5. then $(G\bar{5}, +)$ is a group

Solution :

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$

1) $(G\bar{5}, +)$ ~~G~~ is closed

$\therefore$ all elements in output $\in G$

2) Associative

$$A + (B + C) = (A + B) + C$$

$$\bar{4} + (\bar{3} + \bar{2}) = (\bar{4} + \bar{3}) + \bar{2}$$

$$\bar{4} + \bar{0} = \bar{2} + \bar{2}$$

$$\bar{4} = \bar{4}$$

$\therefore$ $(G\bar{5}, +)$ is associative

3) Identity element $= \bar{0}$

$\therefore$ $\bar{0}$ must be present in every row

4) Inverse

$(\bar{0})^{-1} = \bar{0}$

$(\bar{1})^{-1} = \bar{4}$

$(\bar{2})^{-1} = \bar{3}$

$(\bar{3})^{-1} = \bar{2}$

$(\bar{4})^{-1} = \bar{1}$

Example : 03

let $G_{\bar{5}} = \{\bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ be the set of modulo 5 than $(G_{\bar{5}}, \cdot)$ is a group

Solution

$\cdot$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{2}$	$\bar{2}$	$\bar{4}$	$\bar{1}$	$\bar{3}$
$\bar{3}$	$\bar{3}$	$\bar{1}$	$\bar{4}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{3}$	$\bar{2}$	$\bar{1}$

1) G is closed

2) Associative

$\bar{1} \cdot (\bar{2} \cdot \bar{3}) = (\bar{1} \cdot \bar{2}) \cdot \bar{3}$

$\bar{1} \cdot \bar{1} = \bar{2} \cdot \bar{3}$

$\bar{1} = \bar{1}$

3) identity element $= \bar{1}$

4) inverse

$(\bar{1})^{-1} = \bar{1}$

$(\bar{2})^{-1} = \bar{3}$

$(\bar{3})^{-1} = \bar{2}$

$(\bar{4})^{-1} = \bar{4}$

Example : 04

Show that the set $G = \{\bar{1}, \bar{2}, \bar{4}, \bar{5}, \bar{7}, \bar{8}\}$ under $(\cdot)$ modulo 9 is a group

Solution

$\cdot$	$\bar{1}$	$\bar{2}$	$\bar{4}$	$\bar{5}$	$\bar{7}$	$\bar{8}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{4}$	$\bar{5}$	$\bar{7}$	$\bar{8}$
$\bar{2}$	$\bar{2}$	$\bar{4}$	$\bar{8}$	$\bar{1}$	$\bar{5}$	$\bar{7}$
$\bar{4}$	$\bar{4}$	$\bar{8}$	$\bar{7}$	$\bar{2}$	$\bar{1}$	$\bar{5}$
$\bar{5}$	$\bar{5}$	$\bar{1}$	$\bar{2}$	$\bar{7}$	$\bar{8}$	$\bar{4}$
$\bar{7}$	$\bar{7}$	$\bar{5}$	$\bar{1}$	$\bar{8}$	$\bar{4}$	$\bar{2}$
$\bar{8}$	$\bar{8}$	$\bar{7}$	$\bar{5}$	$\bar{4}$	$\bar{2}$	$\bar{1}$

1) $(G_{\bar{9}}, \cdot)$ is closed

2) Associative

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$
$$\bar{1} \cdot (\bar{2} \cdot \bar{4}) = (\bar{1} \cdot \bar{2}) \cdot \bar{4}$$
$$(\bar{1} \cdot \bar{8}) = (\bar{2}) \cdot \bar{4}$$
$$\bar{8} = \bar{8}$$

3) identity element = 1

4) inverses $(\bar{1})^{-1} = \bar{1}$ $\quad (\bar{7})^{-1} = \bar{4}$

$(\bar{2})^{-1} = \bar{5}$ $\quad (\bar{8})^{-1} = \bar{8}$

$(\bar{4})^{-1} = \bar{7}$

$(\bar{5})^{-1} = \bar{2}$

$(G_{\bar{9}}, \cdot)$ is a group

Example : 05

The set z of all integers under binary operation 'o' defined by $a \circ b = a - b$ is (z, o) is grou

Solution:

Associative property

$A = 1$

$B = 2$

$C = 3$

$$(A \circ B) \circ C = A \circ (B \circ C)$$

$$(1 \circ 2) \circ 3 = 1 \circ (2 \circ 3)$$

$$(-1) \circ 3 = 1 \circ (-1)$$

$$-4 = 2$$

(z, o) is not a group

Theorem :

The only idempotent element in a group is the identity element.

Proof :

let $x \in G$ be idempotent element

$$x^2 = x$$

$$x^{-1} x^2 = x^{-1} x$$

$$x^{-1} x x = x^{-1} x$$

$$e x = e$$

$$\boxed{x = e}$$

Q :

let G be a group show that G is abelion iff $(ab)^2 = a^2 b^2$

Sol :

$$(ab)^2 = \cancel{a(ba)b}\ (ab)(ab)$$

$$= a(ba)b \quad \because \text{Associative}$$

$$= a(ab)b \quad \because \text{abelian}$$

$$= (aa)(bb) \quad \because \text{Associative}$$

$$= a^2 b^2$$

$(ab)^2 = a^2 b^2$

$(ab)(ab) = (aa)(bb)$

$a(ba)b = a(ab)b$ ∴ Associative

$(ba)b = (ab)b$ ∴ left cancellation law

$ba = ab$ ∴ Right cancellation law

G is abelion

clo-1

- integers

clo-2

- logic
- function
- relation
- inclusion
- exclusion
- etc

clo-3

- mathematical induction
- Graph

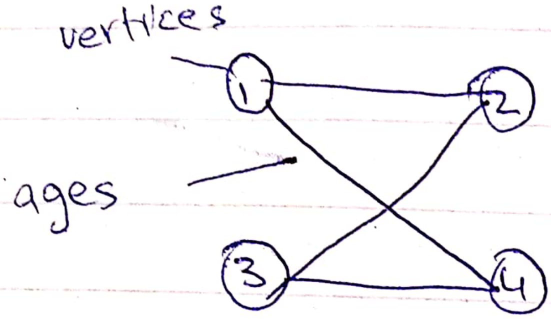
10-01-24 DS Wed

Graphs :

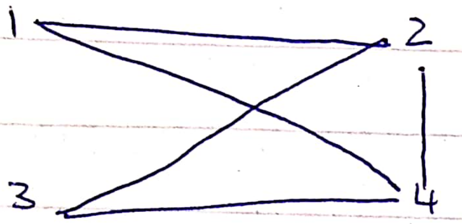
Example : 01 Draw the graph

$V = \{1, 2, 3, 4\}$

$E = \{(1,2) (1,4) (3,4) (2,3)\}$



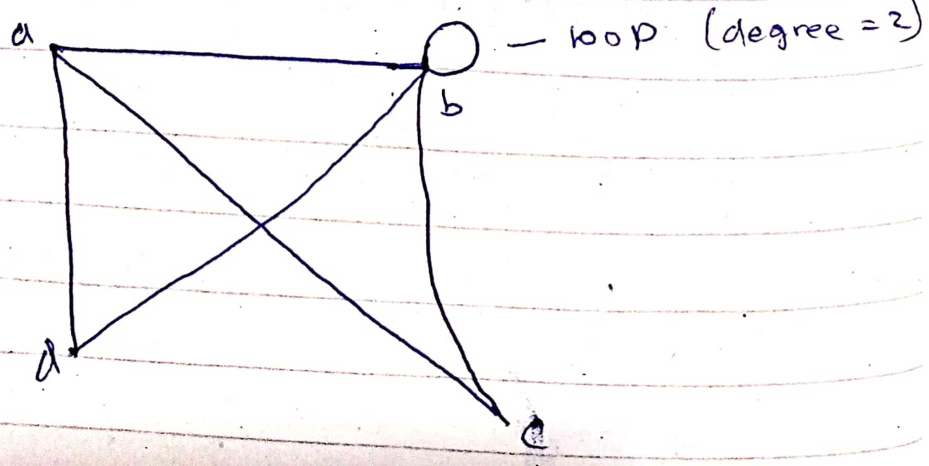
Example : 02 Calculate vertices and ages



$V = \{1, 2, 3, 4\}$

$E = \{(1,2) (1,4) (2,4) (2,3) (3,4)\}$

Degree of a Vertex :



$d(a) = 3$

$d(b) = 5$

$d(c) = 2$

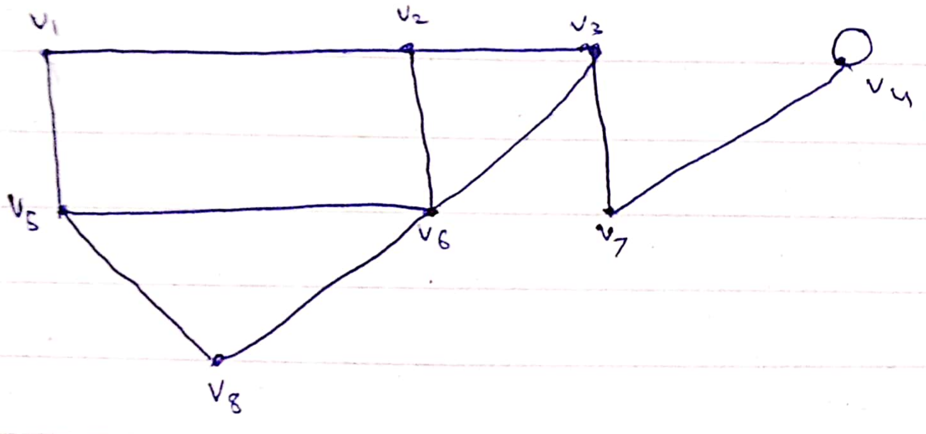
$d(d) = 2$

sum $= 12$

always even

* sum of degress of all the vertices of grap is always even

Q: Calculate the sum of the degree of all vertices.



$d(v_1) = 2$

$d(v_2) = 3$

$d(v_3) = 3$

$d(v_4) = 3$

$d(v_5) = 3$

$d(v_6) = 4$

$d(v_7) = 2$

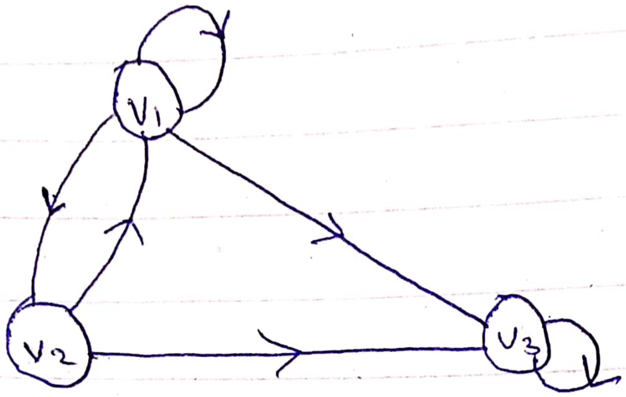
$d(v_8) = 2$

sum $= d(v_1) + d(v_2) + \dots + d(v_8)$

$= 2 + 3 + 3 + 3 + 3 + 4 + 2 + 2$

$= 22$

Indegree & out degree :



$Ind(v_1) = 2$ $\quad$ $Out\,d(v_1) = 3$

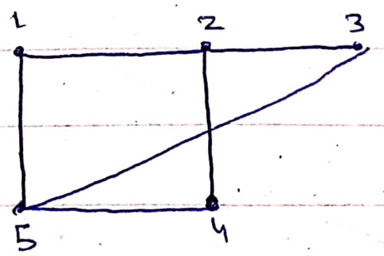
$Ind(v_2) = 1$ $\quad$ $Out\,d(v_2) = 2$

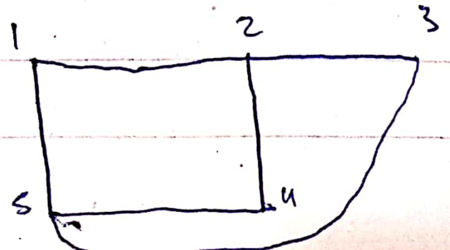
$Ind(v_3) = 3$ $\quad$ $out\,d(v_3) = 1$

Planer Graph

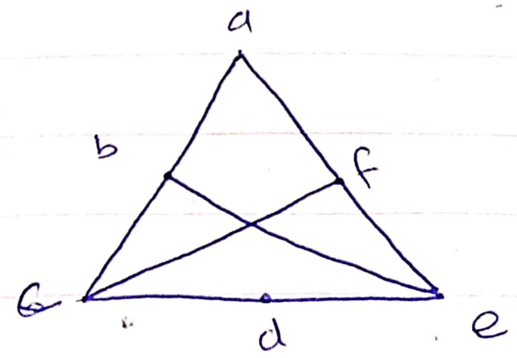
A graph is called planar if it can be drawn in the plane without any edges crossing such a drawing is called a planar graph.

Exp - 1

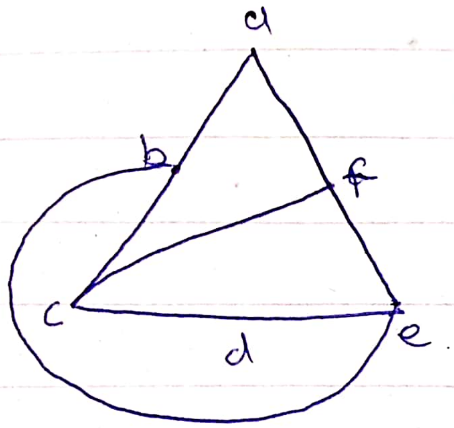




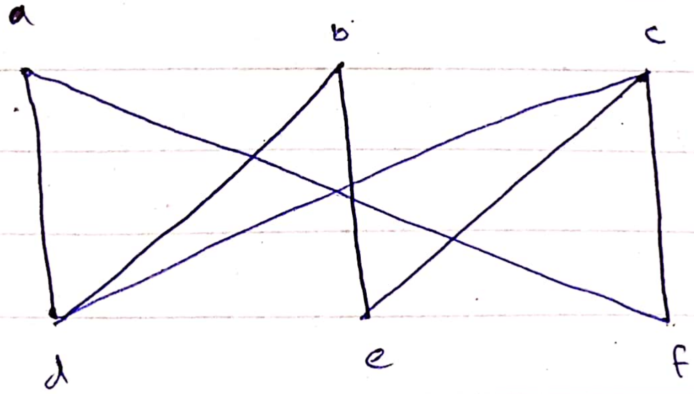
Exp - 2



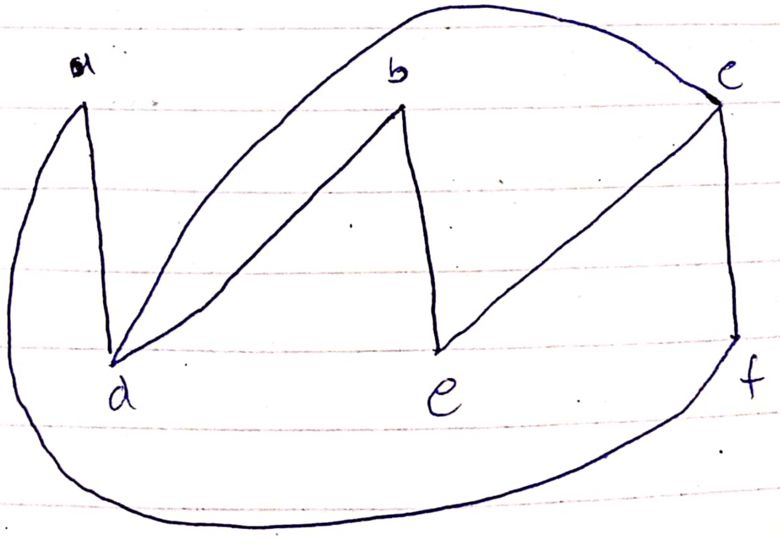
Sol:



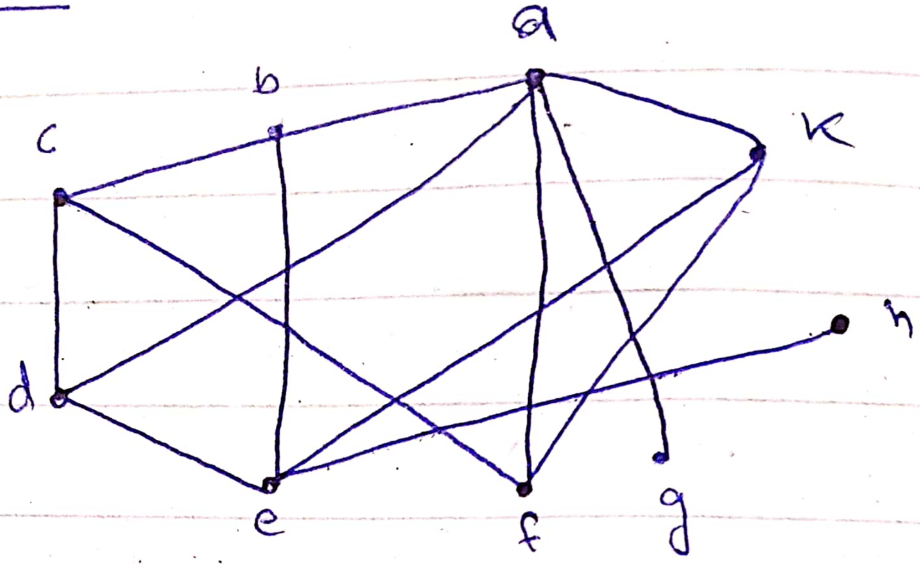
Exp 3



Sol:



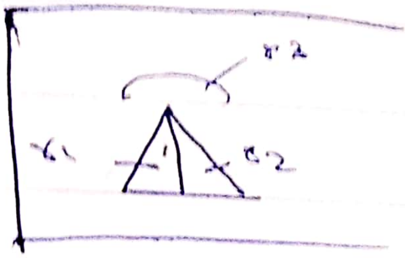
Example - 4



Euler formula:

$$\boxed{r = e - v + 2}$$

r = region
e = edge
v = vertices



Example

let a connected simple planar graph has 20 vertices each of degree 3 Into how many region does a representation of this planar graph split the plane.

Sol:

$$2e = \sum \deg(v)$$

$$2e = 20 \times 3 = 60$$

$$e = 30$$

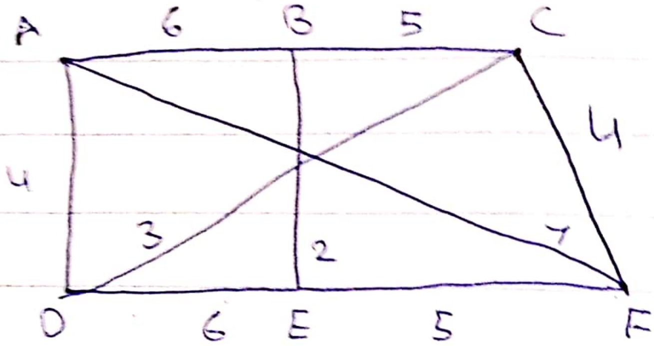
$$r = 30 - 20 + 2$$

$$r = 12$$

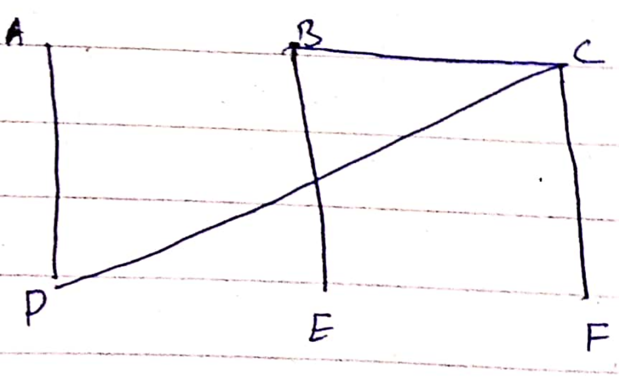
$\longleftrightarrow$

Minimum Spanning Tree :

Using Kruskals Algorithm

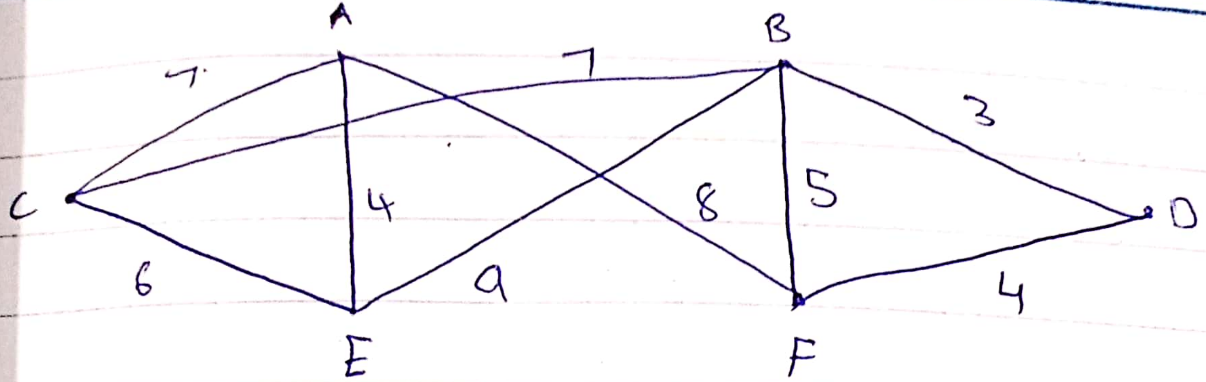


Edges	weights	Added or Not
BE	2	✓
DC	3	✓
AD	4	✓
CF	4	✓
BC	5	✓
EF	5	
AB	6	
DE	6	
AF	7	

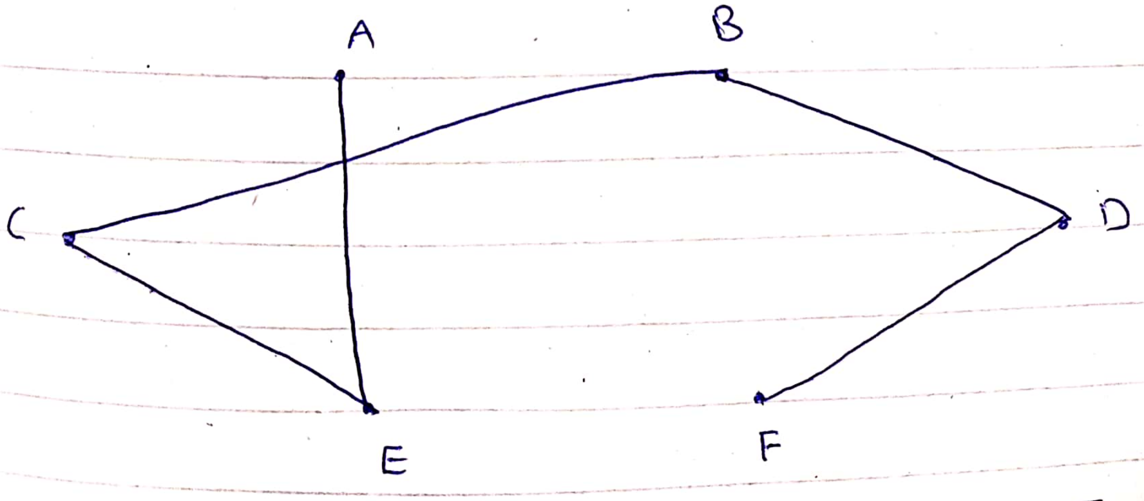


minimum weight $= 2+3+4+4+5$

$= 18$



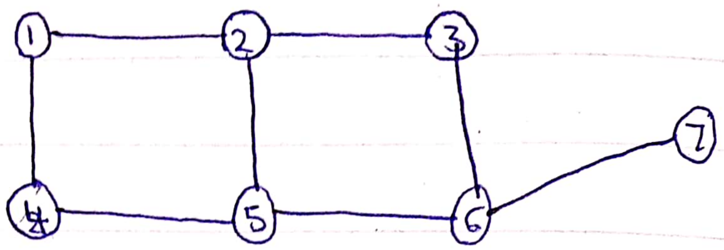
Edgess	weights	Added or Not
BD	3	✓
DF	4	✓
AE	4	✓
BF	5	✗
CE	6	✓
AC	7	✗
BC	7	✓
AF ~~BE~~	8	
BE	9	



minimum weight $= 3 + 4 + 4 + 6 + 7$

$= 24$

Bipartite Graph :

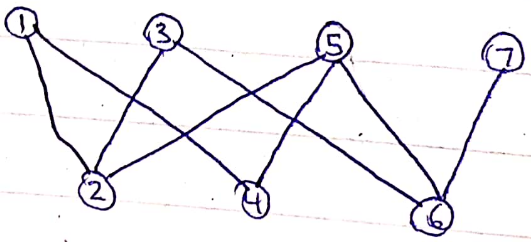


let $1 \in X$

than $2, 4 \in Y$

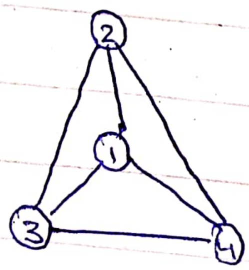
$X = 1, 3, 5, 7$

$Y = 2, 4, 6$



This is a bipartite Graph

Example : 02



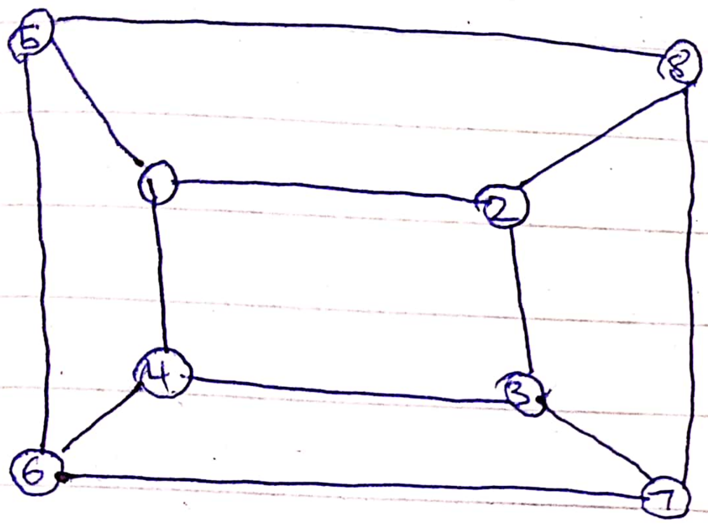
let $1 \in X$

$2, 3, 4 \in Y$

Not possible

$3 \in X$ but it is in Y

Example

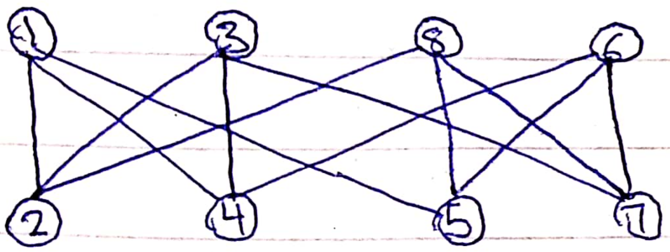


let ~~x~~ $1 \in X$

then $2, 4, 5 \in Y$

$X = 1, 3, 8, 6$

$Y = 2, 4, 5, 7$



This is a bipaptite graph

Mathematical Induction

pdf on 15-

Q: $1^2 + 3^3 + 5^2 + \ldots + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$

Sol:

$n = 0$

$(1)^2 = \frac{(1)(1)(3)}{3} = 1$

$(2k+1)^2 =$

10	clo 1	Integ
15	clo 2	relati
	×	functi
		pigeo
		Inclusion /
	×	counting tec
	×	mathematical
		mathematical induct
25	clo 3	Groups
		Graph